

Theorem (Chebyshev's sum inequality). *Let $n \geq 1$ and let*

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. Then

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \quad (1)$$

Proof. **Step1 – Trivial cases.** If $n = 1$ the statement reduces to $a_1 b_1 \geq a_1 b_1$, which holds with equality. If all a_i are equal (i.e. $a_1 = \cdots = a_n$) then both sides of (1) are equal to $(\sum a_i)(\sum b_i)/n$; the same holds when all b_i are equal. These are the degenerate cases that will later appear as the equality conditions.

Henceforth assume $n \geq 2$ and that the two sequences are not both constant.

Step2 – Centering the sequences. Define the arithmetic means

$$\bar{a} := \frac{1}{n} \sum_{i=1}^n a_i, \quad \bar{b} := \frac{1}{n} \sum_{i=1}^n b_i.$$

Write $a_i = \bar{a} + \alpha_i$ and $b_i = \bar{b} + \beta_i$ where

$$\alpha_i := a_i - \bar{a}, \quad \beta_i := b_i - \bar{b}, \quad \text{so that} \quad \sum_{i=1}^n \alpha_i = \sum_{i=1}^n \beta_i = 0.$$

Then

$$\begin{aligned} n \sum_{i=1}^n a_i b_i &= n \sum_{i=1}^n (\bar{a} + \alpha_i)(\bar{b} + \beta_i) \\ &= n \left(n \bar{a} \bar{b} + \bar{a} \sum_{i=1}^n \beta_i + \bar{b} \sum_{i=1}^n \alpha_i + \sum_{i=1}^n \alpha_i \beta_i \right) \\ &= n^2 \bar{a} \bar{b} + n \sum_{i=1}^n \alpha_i \beta_i. \end{aligned}$$

Since $\sum a_i = n \bar{a}$ and $\sum b_i = n \bar{b}$,

$$\left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) = n^2 \bar{a} \bar{b}.$$

Subtracting gives the identity

$$n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) = n \sum_{i=1}^n \alpha_i \beta_i. \quad (2)$$

Thus (1) is equivalent to

$$\sum_{i=1}^n \alpha_i \beta_i \geq 0. \quad (2')$$

Step3 – Covariance as a double sum. We claim that

$$\sum_{i=1}^n \alpha_i \beta_i = \frac{1}{2n} \sum_{i=1}^n \sum_{j=1}^n (a_i - a_j)(b_i - b_j). \quad (3)$$

Proof of (??). Expanding the right-hand side,

$$\begin{aligned} \frac{1}{2n} \sum_{i,j} (a_i - a_j)(b_i - b_j) &= \frac{1}{2n} \sum_{i,j} (a_i b_i - a_i b_j - a_j b_i + a_j b_j) \\ &= \frac{1}{2n} \left(n \sum_i a_i b_i - \sum_i a_i \sum_j b_j - \sum_j a_j \sum_i b_i + n \sum_j a_j b_j \right) \\ &= \frac{1}{2n} \left(2n \sum_i a_i b_i - 2 \sum_i a_i \sum_j b_j \right) \\ &= \sum_i a_i b_i - \frac{1}{n} \left(\sum_i a_i \right) \left(\sum_j b_j \right). \end{aligned}$$

Since $\bar{a} = \frac{1}{n} \sum a_i$ and $\bar{b} = \frac{1}{n} \sum b_i$, the last line equals

$$\sum_i (a_i - \bar{a})(b_i - \bar{b}) = \sum_i \alpha_i \beta_i.$$

Thus (??) holds. □

Combining (??) and (??) we obtain

$$n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (a_i - a_j)(b_i - b_j). \quad (4)$$

Step4 – Sign of each summand. Because the two sequences are sorted in the same non-decreasing order, for any indices i, j we have

$$i < j \implies a_i \leq a_j \text{ and } b_i \leq b_j,$$

and similarly when $i > j$. Consequently

$$(a_i - a_j)(b_i - b_j) \geq 0 \quad \text{for all } i, j. \quad (5)$$

Each term in the double sum of (??) is non-negative, hence the whole double sum is non-negative. From (??) we deduce

$$n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \geq 0,$$

which is precisely inequality (??).

Step5 – Equality cases. Equality in (??) occurs exactly when the double sum in (??) is zero. By (??) this happens iff every factor $(a_i - a_j)(b_i - b_j)$ is zero. Hence for every pair i, j we must have either $a_i = a_j$ or $b_i = b_j$. The only possibilities compatible with the monotonicity of the two sequences are:

- all a_i are equal (the sequence (a_i) is constant), or
- all b_i are equal (the sequence (b_i) is constant).

The case $n = 1$ is contained in both alternatives. In all other situations the double sum is strictly positive, and the inequality is strict.

Conclusion. For any integer $n \geq 1$ and any two real sequences that are similarly ordered, the Chebyshev sum inequality (??) holds. Equality holds iff at least one of the sequences is constant (including the trivial $n = 1$ case). $\square$ $\square$